Hybrid LiDAR-supervised training with unchanged RGB-only inference

RGB input



single RGB image

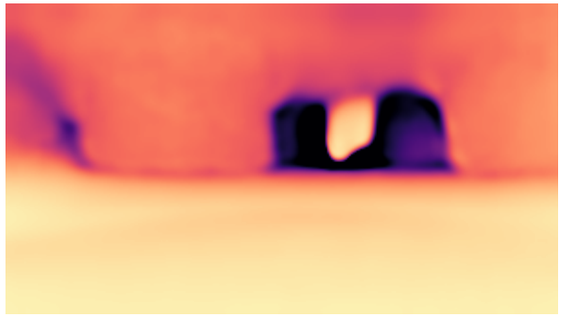
INFERENCE PATH

Lite-Mono

encoder + depth decoder
3.07M inference parameters

depth

Predicted metric depth



loss gradient

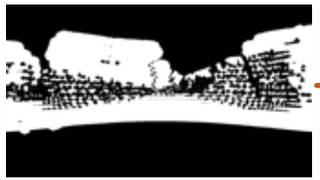
predicted depth

TRAINING ONLY

Projected + local-IDW LiDAR depth



Valid mask



Masked log-L1 loss

valid LiDAR pixels only
weight = 0.1; no scale alignment

Metric supervision

correct absolute scale
during training

At inference, LiDAR, masks, and training losses are removed; the architecture and RGB-only interface remain unchanged.